Calibration-Free Error-Potential Decoding Across Paradigms, and How Good It Must Be to Teach a Robot

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Abstract

Error-related potentials (ErrPs), the brain's response to an observed or committed error, are a long-standing candidate for implicit human feedback to robots. Two questions decide whether this is practical: can a decoder detect ErrPs in a *new* person without a calibration session, and how accurate must it be before its feedback improves learning? We introduce **ErrP-Ens**, a label-free ensemble that combines the within-participant ranks of heterogeneous decoders trained with and without unsupervised per-participant alignment, and **ErrP-Bench**, a cross-subject benchmark of five public ErrP datasets spanning four error paradigms (66 participants, 36,311 epochs), on which we compare ErrP-Ens with a Riemannian pipeline, five compact networks and three EEG foundation models (LaBraM, EEGPT, CBraMod) in 820 cross-subject fits. ErrP-Ens is the best decoder on every dataset, with no training and no labels from the new user, and exceeds the best single decoder of each dataset (chosen with hindsight) on all five. Pretrained foundation models do not outperform compact networks (-0.016 AUROC on average), and the observed agent matters more than any modelling choice: the same participants yield 0.908 AUROC for cursor errors but 0.649 for robot errors. With soft actor-critic on simulated reaching and feedback of controlled quality, we measure a decoder-quality frontier: the benefit of ErrP reward grows with AUROC (Spearman $\rho = 0.58$, $p = 2 \times 10^{-7}$) and turns positive at AUROC 0.77; below it, ErrP feedback slows learning. Cross-subject decoders of cursor, response and orthosis errors are above this point, decoders of observed robot errors are not. Code, data-preparation recipes and per-trial predictions are released.

1 Introduction

When a person watches a machine make a mistake, or makes one themselves, the brain produces an error-related potential (ErrP): a fronto-central negativity followed by a positivity within a few hundred milliseconds of the error [Chavarriaga et al., 2014]. Because ErrPs arise without any explicit action, they have long been proposed as an *implicit* reward signal for teaching robots and interfaces [Iturrate et al., 2015, Kim et al., 2017, Salazar-Gomez et al., 2017, Xu et al., 2021]. Two developments make this question timely. First, EEG *foundation models* pretrained on thousands of hours of recordings [Jiang et al., 2024, Wang et al., 2024, 2025a] promise decoders that transfer to new people without calibration. Second, learning from human feedback has become a central tool in machine learning [Christiano et al., 2017], and a brain signal would make such feedback free.

Whether either promise holds for ErrPs is unclear. Existing EEG foundation-model benchmarks cover many paradigms but at most one ErrP dataset [Lu et al., 2026, Liu et al., 2026a], and ErrP studies typically evaluate a single dataset with its own protocol. On the RL side, studies of ErrP-based reward report gains from simulated or near-ideal feedback, but rarely ask how good a *real* cross-subject decoder must be before its feedback helps.

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We address both questions with a method, a benchmark and a measurement:

- **ErrP-Ens**, a calibration-free decoder (§3). Each participant's EEG is aligned without labels; decoders of three families are trained on other participants with and without alignment; and their outputs are combined by averaging within-participant ranks, which is invariant to the scale and offset differences that make score averaging fail across models and people. ErrP-Ens needs neither training nor labels from the new user.
- **ErrP-Bench**, five public ErrP datasets across four error paradigms: observed robot errors and observed cursor errors [Ehrlich and Cheng, 2019], cursor-monitoring errors [Chavarriaga and Millán, 2010], self-generated response errors (ERN, flanker task) [Kappenman et al., 2021], and errors introduced by an arm orthosis [Kueper et al., 2024]; 66 participants and 36,311 epochs, all used in full under one preprocessing and cross-subject protocol, with a Riemannian pipeline [Rivet et al., 2009, Barachant et al., 2012], five compact networks [Lawhern et al., 2018, Santamaría-Vázquez et al., 2020, Chen et al., 2024a, Song et al., 2023, Zhao et al., 2024] and three foundation models [Jiang et al., 2024, Wang et al., 2024, 2025a], each with and without Euclidean alignment [He and Wu, 2020]; 820 cross-subject fits.
- **A decoder-quality frontier for learning from ErrP feedback**: soft actor-critic [Haarnoja et al., 2018] on FetchReach and FetchPush [Plappert et al., 2018] with feedback of exactly controlled AUROC, which locates the break-even decoder quality at which ErrP reward starts to help, and places every benchmark decoder relative to it.

Findings. (1) ErrP-Ens is the best cross-subject ErrP decoder on all five datasets and is never worse than the best single decoder chosen with hindsight. (2) Pretrained EEG foundation models do not beat compact networks on cross-subject ErrP detection. (3) The observed agent matters more than the decoder: the same participants give 0.908 AUROC for cursor errors and 0.649 for robot errors. (4) Unsupervised alignment helps on some paradigms and hurts on others, which is why ErrP-Ens uses both. (5) On simulated reaching, ErrP feedback helps only above about AUROC 0.77 and slows learning below it, so cross-subject decoders of observed robot errors are not yet on the useful side of the frontier.

All code, data-preparation recipes, per-trial predictions and checkpoints are released.

2 Related work

Error-related potentials. The error-related negativity was first described for self-generated response errors [Falkenstein et al., 1991, Gehring et al., 1993]; observation and interaction ErrPs follow erroneous feedback or machine actions [Chavarriaga and Millán, 2010, Chavarriaga et al., 2014]. Public datasets now span visual observation [Ehrlich and Cheng, 2019, Chavarriaga and Millán, 2010], self-generated errors [Kappenman et al., 2021], physically introduced errors [Kueper et al., 2024, Chari et al., 2024] and mixed sensory feedback [Liu et al., 2026b]. Most decoders are trained within subjects after a calibration session; subject-independent decoders have been built from pooled or simulated participants [Fidêncio et al., 2024], but no common cross-subject protocol compares paradigms, which is the gap ErrP-Bench fills.

ErrPs as implicit feedback for learning. ErrPs have been decoded to correct robot actions online [Salazar-Gomez et al., 2017], to teach interfaces [Iturrate et al., 2015], and to shape the reward of reinforcement learners [Kim et al., 2017, Akinola et al., 2020, Xu et al., 2021]; see Fidêncio et al. [2022] for a review. Recent work aligns robot policies with ErrP-derived reward [Kim et al., 2025], accelerates manipulation learning with ErrP shaping from leave-one-subject-out decoders [Kim et al., 2026], combines ErrPs with adaptive RL interfaces [Fidêncio et al., 2025], and uses EEG-derived reward for driving [Zhuang et al., 2026]. These studies place brain signals in the line of learning from human feedback [Knox and Stone, 2009, Christiano et al., 2017] and report gains for specific decoders, tasks and feedback weights [Kim et al., 2026]. We complement them by measuring the benefit as a continuous function of decoder quality, and by asking which error paradigms reach the required quality without calibration.

EEG decoders and foundation models. Compact convolutional networks such as EEGNet [Lawhern et al., 2018], ShallowConvNet and Deep4Net [Schirrmeister et al., 2017], EEG-

(a) Decoding a new participant u without labels. EEG epochs $X^u_{1:n_u}$ → per-participant Euclidean alignment $\tilde{X}^u_i = \bar{R}_u^{-1/2} X^u_i$ (Eq. 2) → M decoders $f_1, \ldots, f_M$ trained on other participants (with and without alignment) → within-participant rank normalisation $r^u_{m,i}$ (Eq. 5) → average s^u_i (Eq. 6) → error probability $\hat{p}^u_i$ (Eq. 8).

(b) Learning from the decoded signal. Robot segment → participant observes → $\hat{p}_{\text{err}}$ → reward $r_t = r^{\text{task}}_t + w\,(\frac{1}{2} - \hat{p}_{\text{err}})$ (Eq. 9) → soft actor-critic update.

(c) Frontier. Replace $\hat{p}_{\text{err}}$ by feedback of exactly known AUROC (Eq. 10) → learning speed as a function of AUROC → break-even AUROC $A^\star$ (Eq. 11).

Figure 1: Overview of ErrP-Ens and of the decoder-quality frontier.

Inception [Santamaría-Vázquez et al., 2020], EEGNeX [Chen et al., 2024a] and ATCNet [Altaheri et al., 2023], convolution–transformer hybrids [Song et al., 2023, Zhao et al., 2024], and Riemannian pipelines built on xDAWN spatial filtering [Rivet et al., 2009, Barachant et al., 2012, Barachant and Congedo, 2014] are standard for event-related potentials. Foundation models pretrain on large unlabeled corpora and fine-tune on the target task: BENDR [Kostas et al., 2021], BIOT [Yang et al., 2023], LaBraM [Jiang et al., 2024], EEGPT [Wang et al., 2024], CBraMod [Wang et al., 2025a], NeuroLM [Jiang et al., 2025], EEGFormer [Chen et al., 2024b], BrainWave [Yuan et al., 2024], CSBrain [Zhou et al., 2025], REVE [El Ouahidi et al., 2025], LUNA [Döner et al., 2025], CodeBrain [Ma et al., 2026] and LaBraM++ [Barmpas et al., 2025], with state-space alternatives such as FEMBA [Tegon et al., 2025] and EEGMamba [Wang et al., 2025b]. Independent evaluations increasingly find that fine-tuned foundation models gain little over compact networks [Lee et al., 2025] and often fail to beat them when labelled data are scarce [Yang et al., 2026]; our ErrP results agree.

Cross-subject transfer and ensembling. Between-subject shift is reduced without labels by aligning each subject's data, by Euclidean alignment [He and Wu, 2020] or Riemannian recentering [Zanini et al., 2018]. Ensembles of decoders are common in BCI competitions, but averaging raw scores across heterogeneous models requires calibration [Guo et al., 2017], which is unavailable for a new user. ErrP-Ens uses within-participant ranks, which make members commensurable without labels, and includes aligned and unaligned members because the benefit of alignment is paradigm-dependent.

Benchmarks. MOABB [Jayaram and Barachant, 2018, Chevallier et al., 2024] standardised classical BCI paradigms; recent foundation-model benchmarks evaluate many tasks and report mixed gains over specialist models [Kastrati et al., 2025, Xiong et al., 2025, Wu et al., 2025, Lu et al., 2026, Liu et al., 2026a, Kokate et al., 2026]. ErrPs appear there as one task among dozens. ErrP-Bench covers the error paradigms in depth, tests whether alignment, pretraining or ensembling matters most, and measures the downstream value of the decoded signal for learning.

3 Method

Figure 1 summarises the pipeline. A decoder maps one EEG epoch, time-locked to an event that may be an error, to an error score (§3.1). Each participant's data are first aligned without labels (§3.2); a set of decoders is trained on other participants (§3.3); and their scores are combined by ErrP-Ens, a label-free rank ensemble (§3.4). The calibrated score becomes a reward term for a reinforcement learner (§3.5), and a controlled-quality feedback model (§3.6) lets us measure how good the decoder must be for that reward to help.

3.1 Problem: cross-subject error detection

A dataset consists of participants $u \in \mathcal{U}$, each with epochs $X^u_i \in \mathbb{R}^{C \times T}$ (C channels, T samples) and labels $y^u_i \in \{0, 1\}$ (1 = error). A decoder $f_\theta : \mathbb{R}^{C \times T} \to \mathbb{R}$ outputs an error score. In the cross-subject setting the decoder is trained on a set of participants $\mathcal{U}_{\text{train}}$ and applied to a participant $u \notin \mathcal{U}_{\text{train}}$ for whom no labels exist. Its quality for u is the probability that a random error epoch

scores higher than a random correct one,

$$\mathrm{AUROC}_u(f) = \frac{1}{n_1^u n_0^u} \sum_{i:y_i^u=1} \sum_{j:y_j^u=0} \Big(\mathbb{1}[f(X_i^u) > f(X_j^u)] + \tfrac{1}{2}\mathbb{1}[f(X_i^u) = f(X_j^u)] \Big), \quad (1)$$

and the benchmark score is the macro average $\overline{\mathrm{AUROC}} = |\mathcal{U}|^{-1} \sum_u \mathrm{AUROC}_u$ over held-out participants. Because Eq. 1 compares epochs *within* a participant, it is invariant to any strictly increasing per-participant transformation of the scores, a fact ErrP-Ens exploits.

3.2 Unsupervised per-participant alignment

Spatial covariance differs between people (head size, electrode impedance, cortical folding), which shifts the input distribution across participants. Euclidean alignment [He and Wu, 2020] removes the first-order part of this shift using only unlabeled epochs. With band-pass filtered (hence approximately zero-mean) epochs X_i^u,

$$\bar{R}_u = \frac{1}{n_u} \sum_{i=1}^{n_u} \frac{X_i^u X_i^{u\top}}{T}, \qquad \tilde{X}_i^u = \gamma_u \, \bar{R}_u^{-1/2} X_i^u, \qquad \gamma_u = \sqrt{\mathrm{tr}(\bar{R}_u)/C}, \quad (2)$$

so that the aligned mean covariance of every participant is $\gamma_u^2 I_C$. The scale γ_u is our modification: plain EA maps every participant to unit power, which changes the amplitude range that pretrained foundation models were trained on; Eq. 2 preserves each participant's mean channel power while still whitening the spatial structure. Alignment is transductive: it uses all of a test participant's epochs, but never their labels.

3.3 Decoders

We use three decoder families, summarised in Table 1. (i) *Riemannian:* xDAWN spatial filters [Rivet et al., 2009] maximise the signal-to-signal-plus-noise ratio of the class-average responses; the covariance of the filtered epoch augmented with the filtered class templates is projected onto the tangent space of the symmetric positive-definite manifold at the Riemannian mean $\bar{P}$ [Barachant et al., 2012],

$$z_i = \mathrm{upper}\Big(\log\Big(\bar{P}^{-1/2} P_i \bar{P}^{-1/2} \Big) \Big), \qquad f(X_i) = \beta^\top z_i + b, \quad (3)$$

followed by logistic regression. (ii) *Compact networks* trained from scratch: temporal convolution, depthwise spatial convolution and pooling (EEGNet, EEG-Inception, EEGNeX), optionally followed by self-attention over temporal tokens (EEG-Conformer, CTNet). (iii) *Foundation models* pretrained on thousands of hours of EEG with masked reconstruction or masked-token prediction (LaBraM, EEGPT, CBraMod), fine-tuned end to end with a new linear head on the pooled token embeddings. All networks output two logits (ℓ_0, ℓ_1), use the margin $f_\theta(X) = \ell_1 - \ell_0$ as the error score, and minimise class-weighted cross-entropy,

$$\mathcal{L}(\theta) = -\sum_{u \in \mathcal{U}_{\text{train}}} \sum_i \alpha_{y_i^u} \log \mathrm{softmax}\big(\ell_\theta(X_i^u)\big)_{y_i^u}, \qquad \alpha_c = \frac{N}{2N_c}, \quad (4)$$

with early stopping on the macro AUROC of held-out validation participants.

3.4 ErrP-Ens: label-free rank ensembling

Decoders from different families make partly independent errors, but their raw scores live on different scales, and for a new participant there are no labels with which to weight or calibrate them. ErrP-Ens solves both problems with ranks. For member m and participant u, let

$$r_{m,i}^u = \frac{\mathrm{rank}\left(f_m(X_i^u) \,\middle|\, \{f_m(X_j^u)\}_{j=1}^{n_u} \right) - 1}{n_u - 1} \in [0,1], \quad (5)$$

the within-participant rank of epoch i (ties receive average ranks). The ensemble score is the mean rank,

$$s_i^u = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} r_{m,i}^u. \quad (6)$$

Table 1: Decoders evaluated. Params: parameters of the instantiated model for a 27-channel, 2 s input. Pretraining: hours of unlabeled EEG used by the released checkpoint.

Decoder	Family	Core operation	Params	Pretraining
xDAWN+RG [Rivet et al., 2009, Barachant et al., 2012]	Riemannian	spatial filters + tangent space	$<$1k[b]	none
EEGNet [Lawhern et al., 2018]	compact CNN	temporal + depthwise spatial conv.	2.6k	none
EEG-Inception [Santamaría-Vázquez et al., 2020]	compact CNN	multi-scale inception blocks	27.6k	none
EEGNeX [Chen et al., 2024a]	compact CNN	dilated depthwise conv.	56.0k	none
EEG-Conformer [Song et al., 2023]	CNN+Transformer	conv. tokens + self-attention	0.46M	none
CTNet [Zhao et al., 2024]	CNN+Transformer	EEGNet tokens + self-attention	0.15M	none
LaBraM-Base [Jiang et al., 2024]	foundation	masked neural-token prediction	5.8M	$\sim$2,500 h
EEGPT [Wang et al., 2024]	foundation	dual masked self-supervision	25.3M	multi-corpus[a]
CBraMod [Wang et al., 2025a]	foundation	criss-cross attention, masked patches	9.3M	$\sim$27,000 h (TUEG)
ErrP-Ens (ours)	ensemble	rank average of all members, both alignments	0 (no training)	–

[a] several public EEG corpora, see [Wang et al., 2024]. [b] logistic-regression weights on the tangent-space features. Parameter counts are measured on the instantiated models (ErrP-HRI input, 27 channels, 2 s; EEGPT at 250 Hz) and include the new classification head.

Three properties make Eq. 6 suitable for calibration-free use. (1) *No labels:* the ranks use only the participant's unlabeled epochs. (2) *Scale invariance:* $r^u_{m,i}$ is unchanged by any strictly increasing transformation of f_m, so members with heavy-tailed or miscalibrated margins cannot dominate, unlike averaging logits or probabilities. (3) *Per-participant normalisation:* a member whose scores are shifted for a particular participant (for example, by an amplitude difference) still contributes a well-ordered ranking; since AUROC_u depends only on the within-participant order, this normalisation never hurts a single member and removes between-participant offsets from the average.

Members. $\mathcal{M}$ contains every decoder of Table 1 trained *with and without* alignment (one member per decoder–alignment pair, up to 18). Including both is deliberate: alignment helps on some paradigms and hurts on others (§5), and which case applies to a new dataset is unknown without labels; the ensemble hedges between them. We also report two restricted variants: ErrP-Ens-EA (aligned members only) and ErrP-Ens-EA (top-3), which keeps the three members with the highest macro AUROC on the fold's validation participants. A member is included for a training seed only if it finished on every fold, so all test participants of a seed are scored by the same ensemble.

Why averaging ranks helps. If member scores for an error and a correct epoch differ by $\Delta_m = r_{m,i} - r_{m,j}$ with mean $\mu_m > 0$ and covariance Σ across members, the averaged difference has signal-to-noise ratio

$$\mathrm{SNR}(\bar{\Delta}) = \frac{\frac{1}{M}\sum_m \mu_m}{\sqrt{\frac{1}{M^2}\mathbf{1}^\top \Sigma \mathbf{1}}}, \tag{7}$$

which exceeds the best member's $\mu_m/\sqrt{\Sigma_{mm}}$ when members are comparably accurate and their errors are not perfectly correlated. ErrP decoders satisfy both conditions: their AUROCs are within a few points of each other (Table 3) while architectures and alignment settings differ.

From rank to probability. A reward needs a probability rather than a rank. For deployment, the ensemble score can be mapped to an error probability by an isotonic regression g fitted on the validation participants,

$$\hat{p}^u_i = g(s^u_i), \qquad g = \arg\min_{g \text{ non-decreasing}} \sum_{v \in \mathcal{U}_{\text{val}}} \sum_i \left(y^v_i - g(s^v_i)\right)^2, \tag{8}$$

which preserves the ranking, hence the AUROC, and needs no labels from the new participant. All decoding results in this paper are rank-based and do not depend on g; the RL frontier uses the calibrated feedback model of §3.6.

3.5 ErrP feedback as reward

A soft actor-critic agent [Haarnoja et al., 2018] acts in a goal-conditioned task with sparse reward $r^{\text{task}}_t = -\mathbb{1}[\|g - a_t\| > \epsilon]$. Every $k = 5$ steps a segment τ is labelled by the change in task distance d: $y_\tau = 1$ (error) if d increased by more than $\delta = 1$ mm, $y_\tau = 0$ if it decreased by more than δ,

Table 2: ErrP-Bench datasets. "Error" is the label-1 class; epochs are time-locked to the error-eliciting event (feedback, machine action, response or orthosis perturbation).

Dataset	Paradigm	Subj.	Ch.	Epochs	Error %	Min err./subj.	Window (s)
ErrP-HRI robot	observed robot error	12	27	5,822	35.3	155	[0.0, 2.0)
ErrP-HRI cursor	observed cursor error	12	27	5,806	34.2	153	[0.0, 2.0)
BNCI 2015-013	cursor monitoring	6	64	6,437	20.5	188	[0.0, 1.0)
ERP CORE ERN	own response error (flanker)	40	28	15,756	10.6	2	[-0.2, 0.8)
IntEr-HRI	orthosis-introduced error	8	65	2,490	20.0	60	[0.0, 1.0)

and no feedback otherwise. A decoder returns $\hat{p}_\tau$ for the observed segment, and every step of the segment receives

$$r_t = r_t^{\text{task}} + w\left(\tfrac{1}{2} - \hat{p}_\tau\right), \tag{9}$$

with $w = 1$ fixed in advance. With an exact decoder ($\hat{p}_\tau = y_\tau$) the extra term is a $\pm\frac{w}{2}$ progress signal, similar to a coarse potential-based shaping [Ng et al., 1999]; with an uninformative decoder it is noise with a non-zero mean that the critic must fit.

3.6 Feedback of controlled quality and the break-even AUROC

To measure benefit as a function of decoder quality, we replace the EEG decoder by a simulated decoder of exactly known AUROC A. Under the equal-variance binormal model [Green and Swets, 1966], a latent score

$$z_\tau \mid y_\tau \sim \mathcal{N}\big((2y_\tau - 1)\tfrac{d'}{2},\, 1\big), \qquad d' = \sqrt{2}\,\Phi^{-1}(A), \tag{10}$$

has $\Pr(z_\tau > z_{\tau'} \mid y_\tau = 1, y_{\tau'} = 0) = \Phi(d'/\sqrt{2}) = A$. It is converted to the Bayes posterior under the base rate $\pi = 0.35$ on which real decoders are calibrated, $\hat{p}_\tau = \sigma\big(d' z_\tau + \log\frac{\pi}{1-\pi}\big)$, so the simulated decoder is calibrated in the same sense as a real one. For each AUROC level we compute the normalised area under the success curve $\mathrm{AUC} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{succ}(t_k)$ and its paired gain over the sparse-reward run of the same seed, $G(A) = \mathbb{E}_{\text{seed}}[\mathrm{AUC}_A - \mathrm{AUC}_{\text{sparse}}]$. The *break-even AUROC* is the quality at which feedback starts to help,

$$A^\star = \inf\{A :\ G(A) \geq 0\}, \tag{11}$$

estimated by linear interpolation between measured levels. A decoder helps a learner only if its cross-subject AUROC exceeds $A^\star$; comparing the benchmark scores with $A^\star$ tells which paradigms can already provide useful implicit feedback.

4 ErrP-Bench

4.1 Datasets

Table 2 lists the five dataset conditions. All are public and used in full: every participant and every recording of the paradigm. Labels come from each dataset's own event documentation.

ErrP-HRI [Ehrlich and Cheng, 2019]: participants watch a robot (*robot*) or a cursor (*cursor*) execute a direction they selected; in machine-error trials the executed movement differs from the selected one. 12 participants per condition, 27 channels, mastoid reference. We use both conditions as separate datasets, so that the effect of the observed agent is measured with the participants held fixed. **BNCI 2015-013** [Chavarriaga and Millán, 2010]: participants monitor a cursor that moves toward a target and errs in about 20% of steps. **ERP CORE ERN** [Kappenman et al., 2021]: 40 participants perform a flanker task; epochs are locked to the participant's own response, incorrect responses are errors. **IntEr-HRI** [Kueper et al., 2024, Chari et al., 2024]: an active orthosis moves the participant's arm and in some trials briefly reverses (error); correct events are the authors' matched mid-movement markers. Participants squeezed a ball when they noticed an error; in about 90% of error epochs the squeeze falls inside the window (median 0.72 s after the error), so this dataset measures an error-plus-response signal; we report a control restricted to the first 0.5 s.

Preprocessing is identical in kind across datasets: scalp channels only, zero-phase FIR band-pass 1–20 Hz [Gramfort et al., 2013] on the continuous recording, epochs of whole seconds after the event

Table 3: Cross-subject decoding on ErrP-Bench: macro-participant AUROC on held-out participants (mean ± SD over training seeds; *: one seed). EA: Euclidean alignment (✓), none, or both (ErrP-Ens uses members of both kinds). Bold: best per dataset. Rank: mean rank over datasets (†: not run on every dataset). All numbers are measured. Pairwise tests in Appendix A.

Family	Model	EA	ErrP-HRI robot	ErrP-HRI cursor	BNCI 2015-013	ERP CORE ERN	IntEr-HRI	Rank
Riemannian	xDAWN+RG		0.590*	0.823*	0.820*	0.945*	0.955*	15.6
Riemannian	xDAWN+RG	✓	0.603*	0.851*	0.803*	0.950*	0.963*	13.4
compact	EEGNet		0.570±0.008	0.872±0.002	0.823*	0.959±0.002	0.974±0.000	10.0
compact	EEGNet	✓	0.606±0.004	0.885±0.006	0.803±0.002	0.963±0.000	0.947±0.006	11.2
compact	EEG-Inception		0.591±0.000	0.873±0.011	0.849±0.005	0.958±0.002	0.970±0.002	9.2
compact	EEG-Inception	✓	0.632±0.013	0.902±0.002	0.809±0.003	0.966±0.001	0.967±0.002	7.2
compact	EEGNeX		0.623±0.000	0.878±0.010	0.841±0.001	0.963±0.002	0.976±0.003	5.8
compact	EEGNeX	✓	0.622±0.014	0.897±0.005	0.809±0.013	0.968±0.000	0.971±0.004	7.2
compact	EEG-Conformer		0.594±0.004	0.792±0.009	0.820±0.003	0.950±0.001	0.964±0.004	14.0
compact	EEG-Conformer	✓	0.597±0.002	0.840±0.001	0.788±0.000	0.953±0.002	0.960±0.002	14.2
compact	CTNet		0.596±0.016	0.873±0.004	0.846±0.023	0.959±0.001	0.954±0.000	10.8
compact	CTNet	✓	0.617±0.003	0.885±0.005	0.820±0.009	0.961±0.002	0.944±0.006	10.4
foundation	LaBraM		0.554±0.001	0.824±0.002	0.801±0.010	0.944±0.001	0.955±0.007	18.0
foundation	LaBraM	✓	0.564±0.002	0.851±0.015	0.779±0.014	0.952±0.006	0.953±0.007	16.8
foundation	EEGPT		0.582*	0.854*	0.812*	–	0.957*	14.0†
foundation	EEGPT	✓	0.623*	0.891*	0.778*	–	0.920*	13.2†
foundation	CBraMod		0.554±0.008	0.838±0.004	0.819±0.017	0.944±0.002	0.959±0.005	16.2
foundation	CBraMod	✓	0.597±0.001	0.845±0.009	0.823±0.002	0.952±0.000	0.930±0.006	13.8
ours	ErrP-Ens	both	0.642±0.003	0.906±0.002	**0.872±0.004**	0.970±0.001	**0.981±0.001**	1.6
ours	ErrP-Ens-EA	✓	**0.649±0.001**	**0.908±0.003**	0.851±0.002	**0.971±0.001**	0.974±0.002	1.6
ours	ErrP-Ens-EA (top-3)	✓	0.640±0.008	0.905±0.001	0.833±0.001	0.969±0.000	0.973±0.001	4.0

(2 s for ErrP-HRI, the window used in its original analysis; 1 s otherwise), resampling to 256 Hz (Riemannian and compact models), 200 Hz (LaBraM, CBraMod) or 250 Hz (EEGPT), the rates the foundation models were pretrained at, no amplitude-based rejection. Channel names follow the 10–20 system so that channel-aware foundation models receive their own position embeddings.

4.2 Protocol

Cross-subject evaluation is the primary protocol, since an implicit feedback signal is useful only if it works for a new person without a calibration session. Participants of a dataset are split into five disjoint test groups; in each fold, about 15% of the remaining participants form a validation set that is used only for early stopping, and the rest are training participants. Every participant is tested exactly once. The unit of analysis is the held-out participant: we report the macro average over participants of the per-participant AUROC, averaged over training seeds (two seeds per configuration). AUROC is threshold-free and robust to the class imbalance of ErrP data. Differences between decoders are tested with two-sided Wilcoxon signed-rank tests over participants, Holm-corrected within each dataset [Demšar, 2006].

4.3 Training protocol

The decoders of Table 1 are trained on the training participants of each fold. Networks trained from scratch use their braindecode implementations [Schirrmeister et al., 2017], Adam, at most 150 epochs and early stopping with patience 20. Foundation models are initialised from their released pretrained weights and fully fine-tuned with a new classification head (backbone learning rate 10^{-4}, head 5×10^{-4}, AdamW, cosine schedule, at most 30 epochs, patience 10). All networks use class-weighted cross-entropy (Eq. 4) and select the checkpoint with the best validation AUROC. Hyperparameters are the published defaults; no decoder receives dataset-specific tuning, so every decoder has the same tuning budget. Every decoder is trained with and without Euclidean alignment (Eq. 2), with two training seeds (xDAWN is deterministic and run once; EEGPT, the most expensive model, with one seed). ErrP-Ens requires no training: it combines the saved test predictions of the members (Eq. 6).

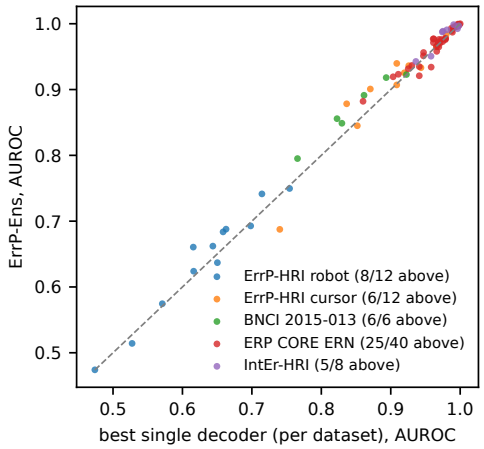


Figure 2: Per held-out participant: ErrP-Ens vs the best single decoder of the same dataset (chosen with hindsight). Points above the diagonal favour ErrP-Ens.

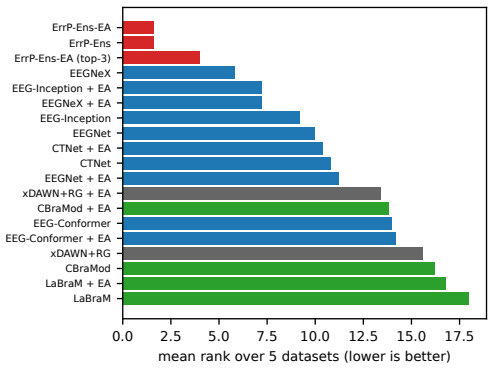


Figure 3: Mean rank over the five datasets (variants run on all datasets). Red: ErrP-Ens; blue: compact networks; green: foundation models; grey: xDAWN.

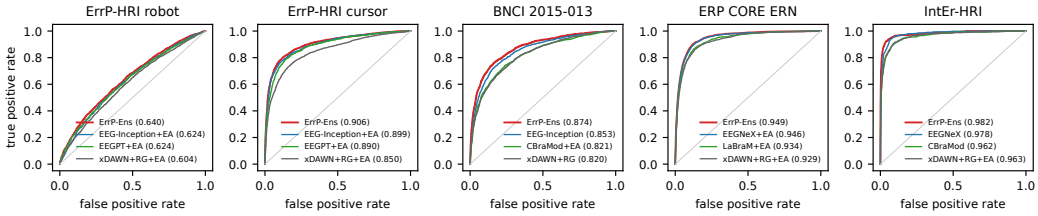


Figure 4: ROC curves on held-out participants (seed 42; scores rank-normalised within participant, as used by ErrP-Ens; AUROC of the pooled curve in brackets) for ErrP-Ens and the best compact, foundation and Riemannian decoder of each dataset.

5 Decoding results

Table 3 reports all 21 decoder variants on the five datasets; Figure 3 summarises them by mean rank.

ErrP-Ens is the most accurate cross-subject ErrP decoder. An ErrP-Ens variant is the best decoder on every dataset, and the headline configuration (all members, both alignment settings) exceeds the best single decoder of each dataset on all five: 0.642 vs 0.632 (EEG-Inception + EA) on ErrP-HRI robot, 0.906 vs 0.902 on ErrP-HRI cursor, 0.872 vs 0.849 on BNCI 2015-013, 0.970 vs 0.968 on ERP CORE ERN and 0.981 vs 0.976 on IntEr-HRI (mean gain +0.009, 5/5 datasets). This comparison is conservative: the best single decoder differs from dataset to dataset and is chosen with hindsight on the test participants, whereas ErrP-Ens uses one fixed rule everywhere. Per participant (Figure 2), the gains are small but consistent; after Holm correction over datasets none of the five differences is individually significant (Appendix A), so we read the result as "never worse, often better" rather than as a large margin. The aligned-only ensemble ErrP-Ens-EA is best on the three datasets where alignment helps (0.649, 0.908, 0.971) and is significantly better than the best single decoder on ERP CORE ERN; selecting members by validation AUROC (top-3) is consistently worse than using all members, which suggests that diversity matters more than the accuracy of individual members.

Decoder outputs. Figure 5 shows the scores of the best foundation model and of ErrP-Ens for the median held-out participant of ErrP-HRI robot. Both separate error from correct trials only partially, as their AUROCs imply; ErrP-Ens shifts error trials toward the top ranks more consistently. The

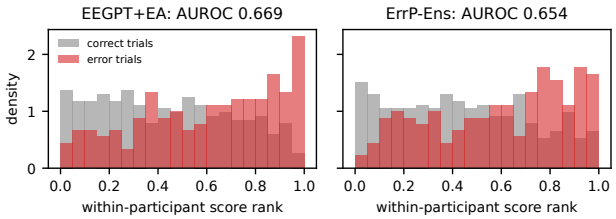


Figure 5: Decoder outputs for one held-out participant (the median participant of ErrP-Ens on ErrP-HRI robot, seed 42): within-participant score ranks of error and correct trials.

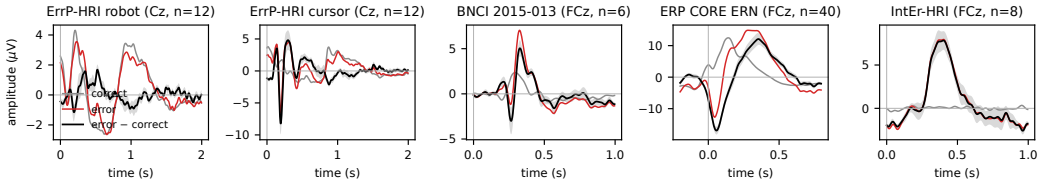


Figure 6: Grand-average ERPs at a fronto-central electrode (mean over participants of the per-participant averages; band: $\pm$ s.e.m. of the error-minus-correct difference).

ROC curves (Figure 4) show that its advantage is spread over the whole operating range rather than concentrated at one threshold.

The observed agent dominates the difficulty. The same twelve participants watched both a robot and a cursor commit the same kind of error. The best decoder reaches 0.908 AUROC on cursor errors but only 0.649 on robot errors. Paired over participants, the cursor condition is higher by 0.259, for 12/12 participants (Wilcoxon $p = 5 \times 10^{-4}$). The grand-average waveforms (Figure 6) show why: the cursor condition produces a sharp, large error-minus-correct deflection, whereas the robot condition produces a small and temporally smeared one. A plausible reason, which we do not test, is that a robot movement unfolds over time, so the moment at which it becomes an error is less sharply defined than a cursor jump.

Pretraining does not help cross-subject ErrP decoding. The best foundation model minus the best compact network averages -0.016 AUROC over datasets, and a foundation model is best on 0/5 datasets; the per-participant difference favours the compact network significantly on ERP CORE ERN and IntEr-HRI (Appendix Table 7). On ErrP-HRI robot, LaBraM and CBraMod are among the weakest decoders, and EEGPT reaches the compact networks only with alignment. One interpretation is that ErrPs are low-dimensional, time-locked waveforms: a model that already encodes spatial filtering and a short temporal template has little left to gain from generic pretraining, whereas a large backbone fine-tuned on a few thousand trials from about ten participants can fit the training participants too closely.

Alignment helps when the paradigm is consistent and hurts otherwise. EA changes AUROC by +0.003 on average over all model–dataset pairs and improves 27/44 of them (Table 4, Figure 7). The sign depends on the dataset: +0.024 on ErrP-HRI cursor (5 of 9 models significantly better) and +0.023 on ErrP-HRI robot, but -0.024 on BNCI 2015-013 and -0.012 on IntEr-HRI (3 models significantly worse). Our hypothesis for the negative cases is that when the spatial covariance is shaped by the task itself (in IntEr-HRI, the arm movement and the ball squeeze), whitening it also removes signal. Because the sign cannot be known for a new paradigm without labels, ErrP-Ens includes both aligned and unaligned members, and it is the only configuration that is best or near-best on every dataset.

IntEr-HRI measures more than an ErrP. Decoders reach AUROC near 0.981 on IntEr-HRI. In about 90% of its error epochs, the participant's ball squeeze (a motor response to the error) falls inside the window. Restricted to the first 0.5 s, before almost all squeezes (0.8% occur earlier),

Table 4: Effect of Euclidean alignment: AUROC with EA minus without, averaged over seeds and participants. *: paired Wilcoxon $p < 0.05$ over participants (uncorrected).

Model	ErrP-HRI robot	ErrP-HRI cursor	BNCI 2015-013	ERP CORE ERN	IntEr-HRI	Mean Δ
xDAWN+RG	+0.013	+0.028*	-0.017	+0.006	+0.008	+0.007
EEGNet	+0.036*	+0.013	-0.020	+0.005*	-0.027*	+0.002
EEG-Inception	+0.041	+0.028*	-0.041*	+0.009*	-0.003	+0.007
EEGNeX	-0.001	+0.018	-0.033	+0.006*	-0.005	-0.003
EEG-Conformer	+0.002	+0.048*	-0.032	+0.002	-0.004	+0.004
CTNet	+0.021*	+0.012	-0.026	+0.003	-0.010	+0.000
LaBraM	+0.009	+0.026*	-0.021	+0.008	-0.002	+0.004
EEGPT	+0.041*	+0.037*	-0.034	–	-0.037*	+0.002
CBraMod	+0.043*	+0.007	+0.004	+0.007	-0.029*	+0.006

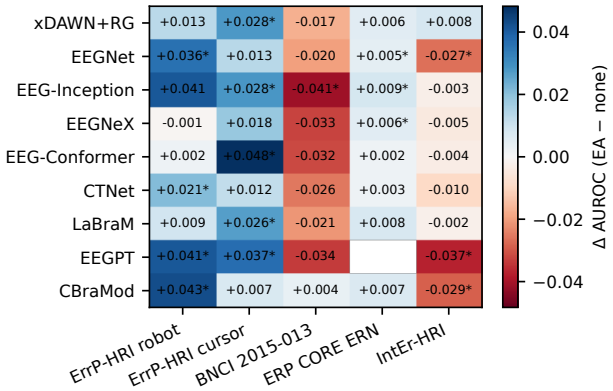


Figure 7: EA effect per decoder and dataset (*: $p < 0.05$, uncorrected).

EEGNet still reaches 0.934 and xDAWN 0.934. Most of the discriminative signal therefore precedes the squeeze, but the full-window scores should be read as an upper bound for an error-plus-response signal.

6 How good must an error decoder be? A decoder-quality frontier for learning

Decoding accuracy is a proxy. The quantity that matters for implicit feedback is whether a learner improves when its reward includes the decoded error probability. We measure this directly, as a function of decoder quality, with real reinforcement learning.

Setup. Soft actor-critic [Haarnoja et al., 2018] (Stable-Baselines3 [Raffin et al., 2021]) learns FetchReach and FetchPush (Gymnasium-Robotics [de Lazcano et al., 2024, Plappert et al., 2018], MuJoCo [Todorov et al., 2012]) from the sparse task reward plus ErrP feedback. Every five simulator steps, the change in the task distance (gripper–goal for Reach, object–goal for Push) labels the segment: an increase beyond 1 mm is an error, a decrease beyond 1 mm a correct step, anything else yields no feedback. For a labelled segment with label y, a decoder reports p_{err} and the reward receives $w\,(0.5 - p_{\text{err}})$ with $w = 1$, the value frozen in our earlier validation study. Budgets are 150k environment steps (Reach) and 500k (Push); evaluation uses 100 fixed goals, a deterministic policy and the sparse success criterion every 10k steps. Learning speed is the normalized area under the success curve (AUC), the primary metric because final success saturates on Reach.

Decoders of controlled quality. To trace the frontier we replace the EEG decoder with a decoder of exactly known AUROC. Under the equal-variance binormal model [Green and Swets, 1966, Hanley and McNeil, 1982], a margin $z \sim \mathcal{N}(\pm d'/2, 1)$ with $d' = \sqrt{2}\,\Phi^{-1}(\text{AUROC})$ has the target AUROC; it is turned into a Bayes-calibrated probability with base rate $\pi = 0.35$, the error prevalence on which the real decoders are calibrated. AUROC levels are 0.55, 0.60, 0.65, 0.70, 0.80,

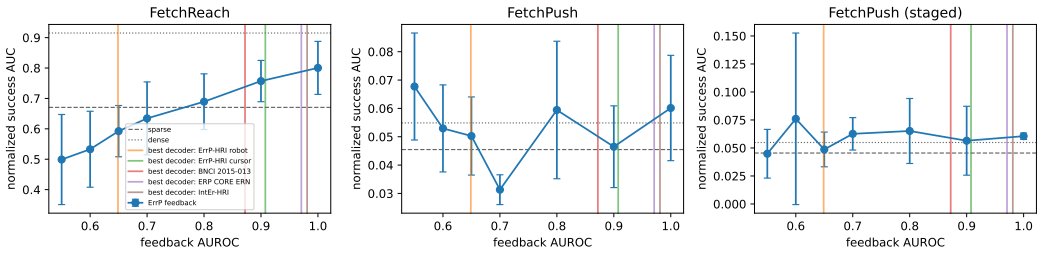


Figure 8: Decoder-quality frontier. Normalized success AUC (mean ± SD over five seeds) of SAC with ErrP feedback of controlled AUROC; horizontal lines: sparse and dense reward. Vertical lines: the cross-subject AUROC of the best ErrP-Bench decoder on each dataset.

Table 5: Normalized success AUC (mean ± SD over seeds) and its paired difference to the sparse-reward run of the same seed (k/n: seeds with a positive difference).

Feedback	FetchReach		FetchPush		FetchPush (staged)	
	norm. AUC	Δ vs sparse	norm. AUC	Δ vs sparse	norm. AUC	Δ vs sparse
sparse only	0.671±0.106	ref	0.045±0.011	ref	0.045±0.011	ref
dense (reference)	0.915±0.029	+0.245 (10/10)	0.055±0.007	+0.009 (5/5)	0.053±0.015	+0.008 (4/5)
AUROC 0.55	0.499±0.148	-0.172 (1/10)	0.068±0.019	+0.022 (4/5)	0.045±0.022	-0.001 (2/5)
AUROC 0.60	0.533±0.125	-0.138 (2/10)	0.053±0.015	+0.007 (4/5)	0.076±0.077	+0.031 (3/5)
AUROC 0.65	0.592±0.084	-0.078 (1/10)	0.050±0.014	+0.005 (4/5)	0.049±0.015	+0.003 (3/5)
AUROC 0.70	0.634±0.120	-0.036 (5/10)	0.031±0.005	-0.014 (0/5)	0.063±0.014	+0.017 (5/5)
AUROC 0.80	0.689±0.092	+0.019 (5/10)	0.059±0.024	+0.014 (4/5)	0.065±0.029	+0.020 (4/5)
AUROC 0.90	0.757±0.068	+0.087 (6/10)	0.046±0.014	+0.001 (2/5)	0.056±0.031	+0.011 (3/5)
AUROC 1.00 (exact)	0.800±0.087	+0.130 (9/10)	0.060±0.019	+0.015 (4/5)	0.061±0.003	+0.015 (4/5)

0.90 and 1.0 (exact labels), with sparse and dense (10× distance decrease) reward as references. Seeds per condition: 10 (FetchReach), 5 (FetchPush), 5 (FetchPush (staged)). Because the object rarely moves early in FetchPush, the object–goal rule yields little feedback there; a supplementary *staged* variant labels segments by the change in gripper–object plus object–goal distance (its sparse reference is the same sparse run, since the rule does not affect sparse training).

Results. On FetchReach (Figures 8 and 9, Table 5), the benefit of ErrP feedback rises monotonically with decoder quality: across all runs, the paired gain in normalized success AUC over the sparse run of the same seed is rank-correlated with feedback AUROC (Spearman $\rho = 0.58$, $p = 2 \times 10^{-7}$). Exact feedback raises the AUC from 0.671 to 0.800 (9/10 seeds better, sign test $p = 0.021$), about half of the gain of the dense reward (0.915). The mean gain crosses zero at AUROC 0.77 (linear interpolation between the measured levels). Below this point the feedback is harmful: at AUROC 0.55 the AUC drops to 0.499 (-0.172, 1/10 seeds better, $p = 0.021$). Final success saturates for every condition (at least 0.92), so the effect is on learning speed. Noisy feedback is therefore not merely uninformative; a likely reason is that, with a fixed weight, it adds a reward term that is only weakly correlated with progress and that the critic must fit nonetheless.

On FetchPush, SAC without hindsight relabelling does not learn the task within 500k steps in any condition (final success 0.04 with sparse reward, 0.05 with dense reward, 0.09 with exact ErrP feedback), and no condition differs reliably from sparse. Push therefore shows that ErrP feedback of any quality does not substitute for exploration when the learner rarely moves the object; it does not locate a frontier. The staged label rule, which provides several times more feedback events, gives the same picture (Table 5, right).

Placing the benchmark on the Reach frontier (vertical lines in Figure 8): the best cross-subject decoders of cursor errors, cursor-monitoring errors, response errors and orthosis errors lie above the break-even point, whereas the best decoder of observed robot errors (0.649) lies well below it, in the range where feedback slowed learning.

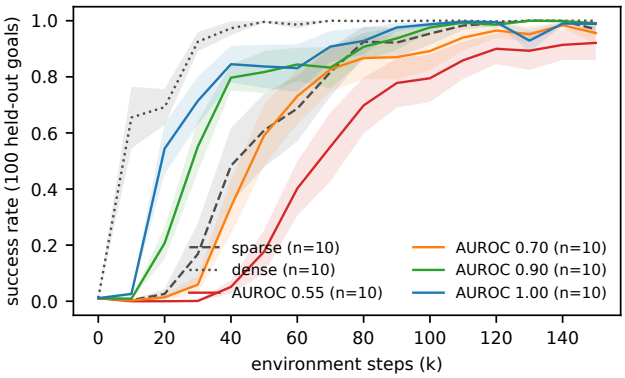


Figure 9: FetchReach learning curves (success on 100 held-out goals; mean $\pm$ s.e.m. over 10 seeds) for sparse and dense reward and for ErrP feedback of four quality levels. Low-quality feedback delays learning relative to the sparse reward; high-quality feedback accelerates it.

7 Discussion and limitations

Why a simple ensemble wins. Decoders of different families reach similar AUROC but disagree on individual trials, and alignment helps some of them on some datasets and hurts on others. Averaging within-participant ranks exploits both facts without labels: it removes the scale and offset differences that prevent averaging raw scores, and it lets aligned and unaligned members hedge against each other. The gain over the best single decoder is modest in absolute terms, but it is obtained without choosing that decoder, which in practice cannot be done for a new paradigm or user. ErrP-Ens was defined after the alignment results were known (Section 5); its aligned-only predecessor, fixed before any result, is reported alongside.

What the benchmark says about implicit feedback. Two numbers have to meet for ErrP feedback to help a learner: the AUROC a decoder reaches for a new person, and the AUROC the learner needs. ErrP-Bench measures the first across paradigms; the frontier measures the second. For observed robot errors, the paradigm that motivates the field, they do not meet in our study: the best cross-subject decoder (0.649) is far below the break-even level on reaching (0.77), and feedback at that quality slowed learning relative to the sparse task reward. This agrees with a preliminary experiment of ours in which feedback from a real cross-subject decoder of robot errors gave no gain over the sparse reward on the same task. It suggests that progress will come from paradigms and interfaces that make errors more salient to the observer, or from per-user calibration, rather than from larger pretrained decoders.

Limitations. (i) The frontier uses synthetic feedback of controlled AUROC under an equal-variance binormal model with independent noise per segment; real decoder errors are correlated across trials and participants, which could move the break-even point. (ii) Two simulated tasks and one RL algorithm with a fixed feedback weight ($w = 1$, frozen beforehand) are studied; a learner that discounts unreliable feedback could lower the break-even AUROC. (iii) We did not tune hyperparameters for any decoder; tuning could favour the larger models, although it would also break the equal-budget comparison. (iv) The foundation-model results depend on our fine-tuning recipe (full fine-tuning, a new linear head, a short schedule). (v) EEGPT, the slowest model, was run with one seed and is missing on ERP CORE ERN (Table 3). (vi) In IntEr-HRI, a motor response falls inside most error windows; the early-window control shows that much, but not necessarily all, of the signal precedes it. (vii) Cross-subject evaluation is our primary protocol; within-subject calibration would reach higher AUROC and is not covered by the frontier placement. (viii) The gains of ErrP-Ens over the best single decoder are small and not individually significant after multiple-comparison correction; the case for it rests on consistency across datasets.

Ethics and data. All datasets are public recordings collected by their authors under their own ethics approvals; we redistribute no EEG data, only preparation code. The ErrP-HRI dataset declares

no license; the others are released under Creative Commons licenses (BNCI 2015-013: CC BY-NC-ND 4.0; ERP CORE: CC BY-SA 4.0; IntEr-HRI: CC BY 4.0).

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A Statistical comparisons

For each dataset, the best variant of Table 3 is compared with every other complete variant by a two-sided Wilcoxon signed-rank test over participants (seed-averaged per-participant AUROC), Holm-corrected within the dataset. With 6 (BNCI 2015-013) or 8 (IntEr-HRI) participants the smallest attainable two-sided p is 0.031 and 0.008; after Holm correction over about twenty comparisons no difference on these two datasets can reach $p < 0.05$, so non-significance there is not evidence of equivalence.

B Data preparation details

ErrP-HRI [Ehrlich and Cheng, 2019]: robot and cursor sessions of the same 12 participants; epochs $[0, 2)$ s after the machine action; mastoid reference as recorded. **BNCI 2015-013**: 6 participants, 64 channels, common average reference; event codes 5/10 (correct) and 6/9 (error), epochs $[0, 1)$ s. **ERP CORE ERN**: 40 participants, P9/P10 reference as in the ERP CORE pipeline, response-locked epochs $[-0.2, 0.8)$ s; one participant has only two error trials and is kept. **IntEr-HRI**: 8 participants, 65 channels, common average reference with FCz restored; error marker S96, correct marker S48, epochs $[0, 1)$ s; the early-window control uses $[0, 0.5)$ s at 256 Hz. All filtering is applied to the continuous recording before epoching. Prepared arrays are validated (shapes, labels, per-participant counts, absence of NaNs) and a within-dataset LDA sanity check is run before any model is trained.

Table 6: Best variant per dataset versus all others.

Dataset	Best variant	Sig. worse / others	Not significantly different (Holm $p \geq 0.05$)
ErrP-HRI robot	ErrP-Ens-EA	18/20	EEGPT+EA, ErrP-Ens
ErrP-HRI cursor	ErrP-Ens-EA	17/20	EEG-Inception+EA, ErrP-Ens, ErrP-Ens-EA (top-3)
BNCI 2015-013	ErrP-Ens	0/20	CBraMod+EA, CBraMod, CTNet+EA, CTNet, EEG-Conformer+EA, EEG-Conformer, EEG-Inception+EA, EEG-Inception, EEGNet+EA, EEGNet, EEGNeX+EA, EEGNeX, EEGPT+EA, EEGPT, ErrP-Ens-EA, ErrP-Ens-EA (top-3), LaBraM+EA, LaBraM, xDAWN+RG+EA, xDAWN+RG
ERP CORE ERN	ErrP-Ens-EA	17/18	ErrP-Ens
IntEr-HRI	ErrP-Ens	0/20	CBraMod+EA, CBraMod, CTNet+EA, CTNet, EEG-Conformer+EA, EEG-Conformer, EEG-Inception+EA, EEG-Inception, EEGNet+EA, EEGNet, EEGNeX+EA, EEGNeX, EEGPT+EA, EEGPT, ErrP-Ens-EA, ErrP-Ens-EA (top-3), LaBraM+EA, LaBraM, xDAWN+RG+EA, xDAWN+RG

Table 7: Best foundation model versus best compact network per dataset (paired Wilcoxon over participants, uncorrected).

Dataset	Best foundation	AUROC	Best compact	AUROC	Δ	FM higher	p
ErrP-HRI robot	EEGPT + EA	0.623	EEG-Inception + EA	0.632	-0.009	6/12	0.677
ErrP-HRI cursor	EEGPT + EA	0.891	EEG-Inception + EA	0.902	-0.011	3/12	0.110
BNCI 2015-013	CBraMod + EA	0.823	EEG-Inception	0.849	-0.027	1/6	0.062
ERP CORE ERN	LaBraM + EA	0.952	EEGNeX + EA	0.968	-0.016	5/40	8×10^{-8}
IntEr-HRI	CBraMod	0.959	EEGNeX	0.976	-0.017	2/8	0.039

C Compute and coverage

All decoders were trained on NVIDIA A100 GPUs (xDAWN on CPU); the RL frontier used one CPU core per run. The benchmark comprises 820 cross-subject fits (about 95 GPU-hours). EEGPT is about an order of magnitude slower per epoch than the other foundation models; within the compute window of this study it was run with fewer seeds, and some of its alignment conditions are missing (marked in Table 3). xDAWN is deterministic and was run once per configuration; it uses Ledoit–Wolf shrinkage both for its spatial filters and for the covariances, because the unregularised filter estimate is singular on the common-average-referenced datasets. A complete per-variant listing (seeds, participants, completeness) is in the released `results_summary.txt`. Partially finished seeds are excluded, never imputed.

D Reproducibility

Every number in this paper is written by `bench/make_paper.py` from the released result files; no number is typed by hand. Fold assignments, configurations and per-trial test predictions of every fit are released, so that every table can be recomputed without retraining.